1- consider the matrix $A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ some of the elimination matrices are given below:

a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{bmatrix}$$

k, m

$kx_2+x_4=0$

$x_2=x_3$ $\quad -\frac{1}{2}x_3+x_4=0$ $\quad k=-\frac{1}{2}$

$mx_1-x_3=0$ $\quad m=-8$

$m x_1=-x_3$ $\quad -8x_1+x_2=0$

$x_2=8x_1$

So, $x_2=x_3$ and $x_2=8x_1$ and

$-mx_1=x_3$

$-mx_1=8x_1$

$m=-8$

c)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -\frac{1}{2} & 0 & 1 \end{bmatrix} = 1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ -8 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} = 1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{bmatrix} = 1$$

3- Suppose u is a unit vector in R^n. This problem is about the n by n symmetric matrix $H = I - 2uu^T$:

b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T).(I - 2(u^T)^T u^T)$

$= (I - 2uu^T).(I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T.uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$

$= I - 4uu^T + 4uu^T$

$= I - 4(+) + 4(+)$

$H.H^T = I$

$H^T H = I$

So, $HH^T = H^T.H = I$

& 'H' is an Orthogonal matrix

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

a) $H^2 = H.H = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - I.2uu^T - 2uu^T I + 4uu^T.uu^T$

$= I(I - 2uu^T - 2uu^T) + 4u.\|u\|^2.u^T$

$= I - 2uu^T - 2uu^T + 4u.\|u\|^2.u^T$

$\|u\|^2 = u^T.u$

$\|u\| = \pm\sqrt{u^T u}$

$\|u\| = 1$

$H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4(1) + 4(1)$

$\rightarrow \boxed{H^2 = I}$

d) Given, $H = (I - 2uu^T)$

$$H^T = (I - 2uu^T)^T$$
$$= I^T - 2(uu^T)^T$$
$$= I - 2(u^T)^T u^T$$
$$= I - 2uu^T = H$$

$(A-B)^T = A^T - B^T$

$(AB)^T = B^T A^T$

H symmetric matriksdir.

$$H = Q \Lambda Q^T$$
$$= Q \Lambda Q^{-1} \quad (Q \text{ is an orthogonal matrix})$$
$$= (Q^{-1})^{-1} \Lambda Q^{-1} \quad (\Lambda \text{ diagonal matrix})$$
$$= P^{-1} \Lambda P$$

H diagonalizable matriksdir.

c) For a matrix A, a vector (non zero will be eigen-vector (suppose, v) of A corresponding eigenvalue λ if, $\boxed{Av = \lambda v}$

$$Hu = (I - 2uu^T) \cdot u$$
$$= Iu - 2uu^T u$$
$$= u - 2u \cdot \|u\|^2$$

$$Iu = 1 \cdot u = u$$

$$\|u\| = +\sqrt{u^T u}$$

$$Hu = u - 2u \cdot 1 = -u$$
$$Hu = -u = (-1)u$$
$$Hu = u - 2u \cdot 1 = -u$$
$$Hu = -u = (-1)u$$
$$Hu = \lambda u \quad \lambda = -1$$

u is eigen vector

$u = 1$

On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

Mustafa Taha Kabar

